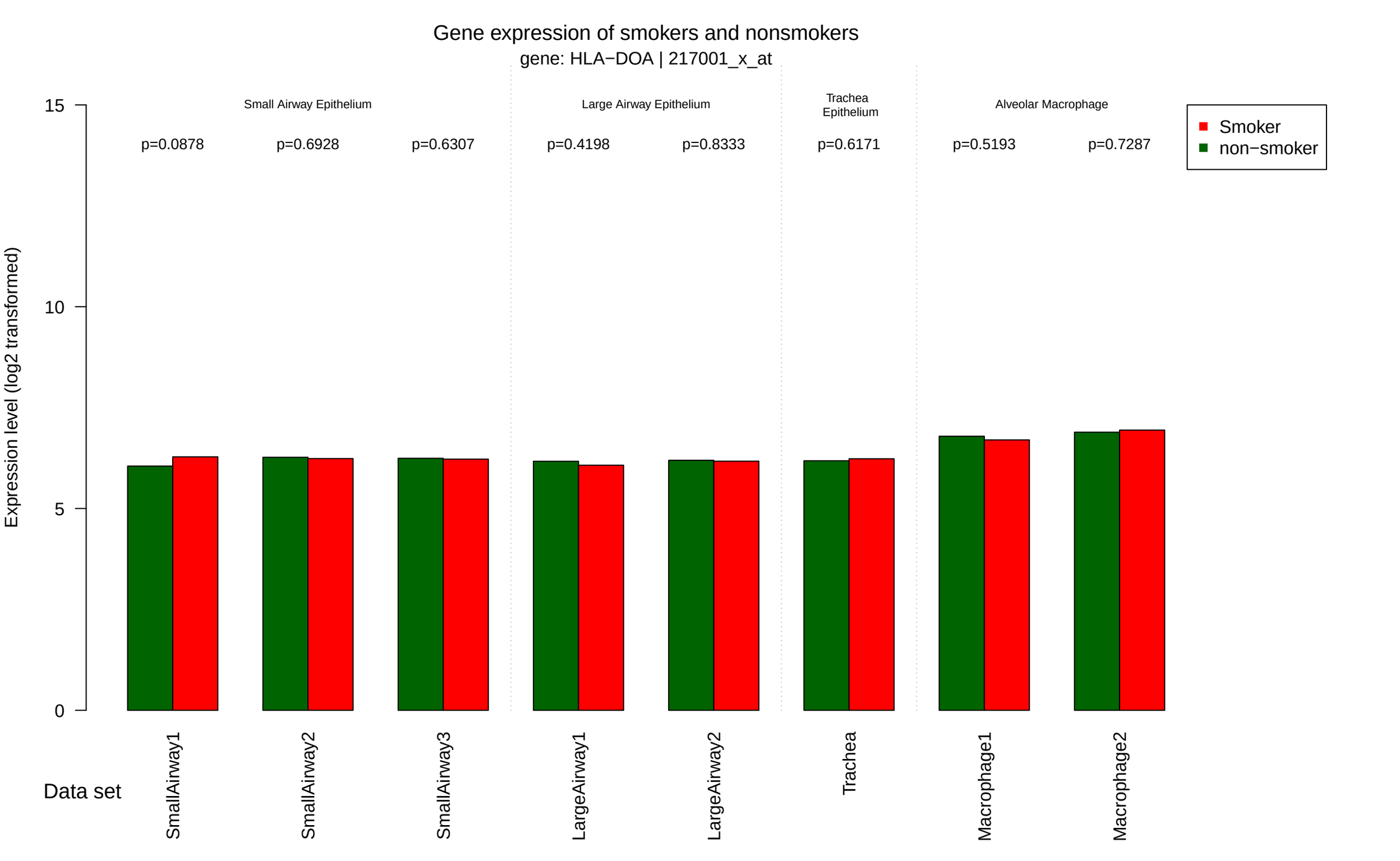
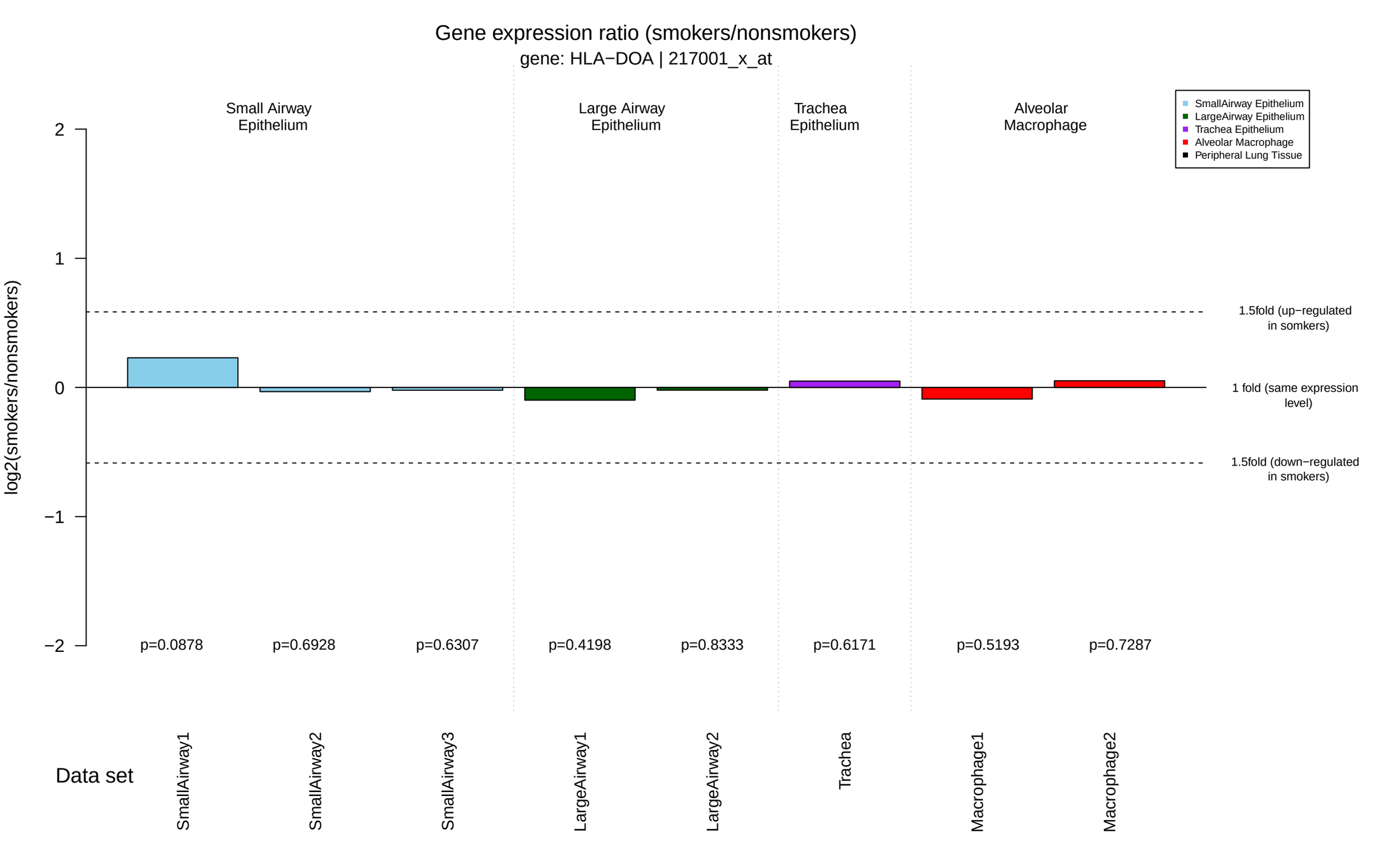
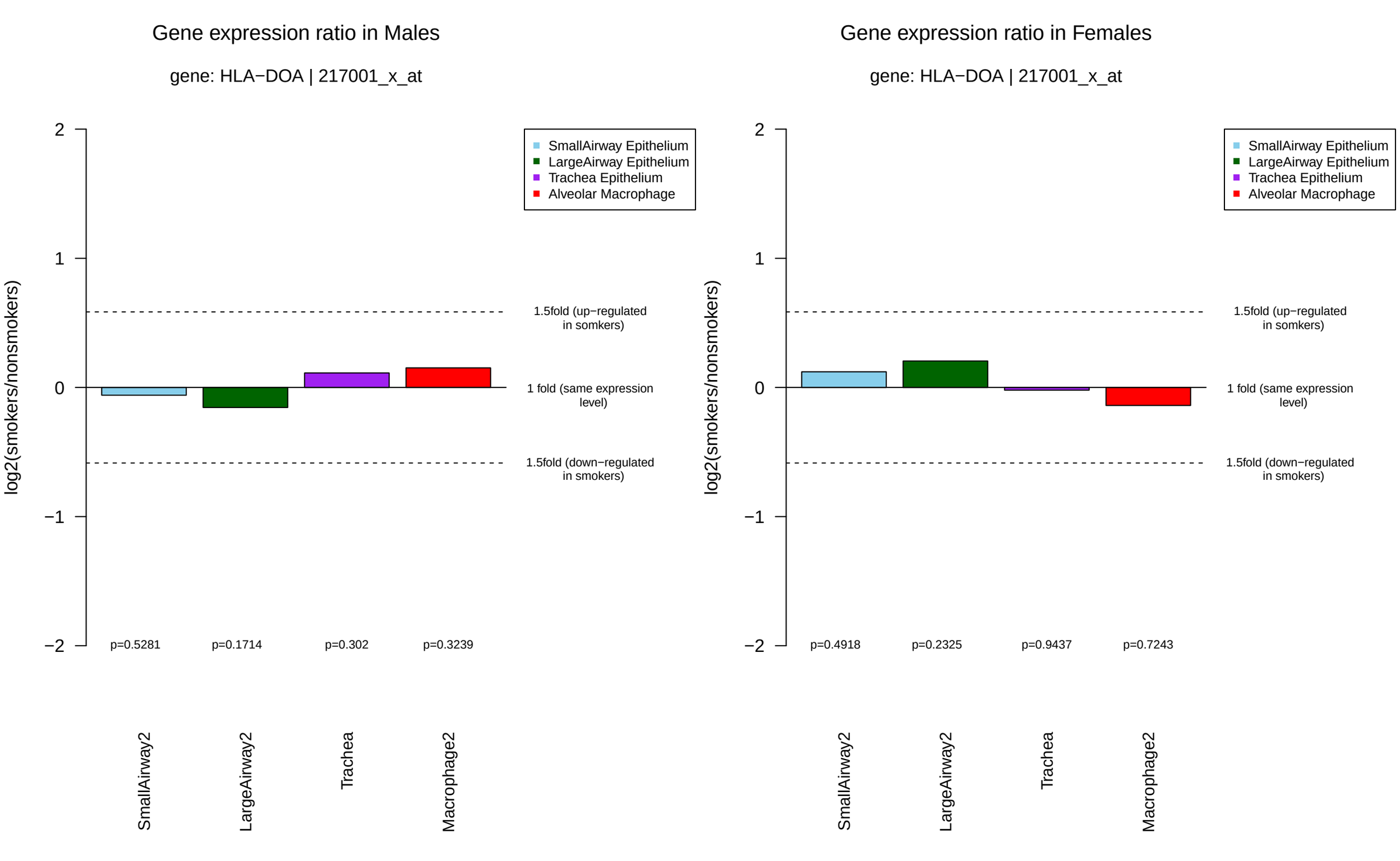
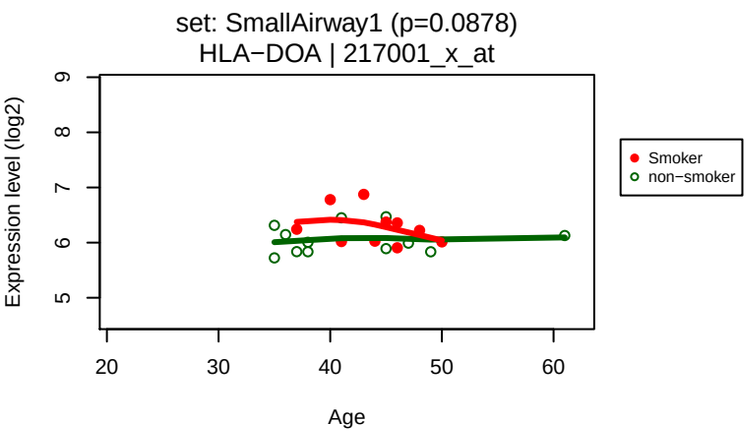
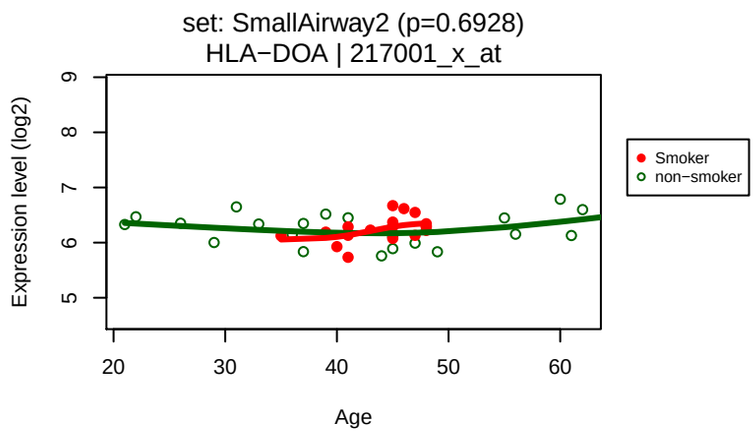
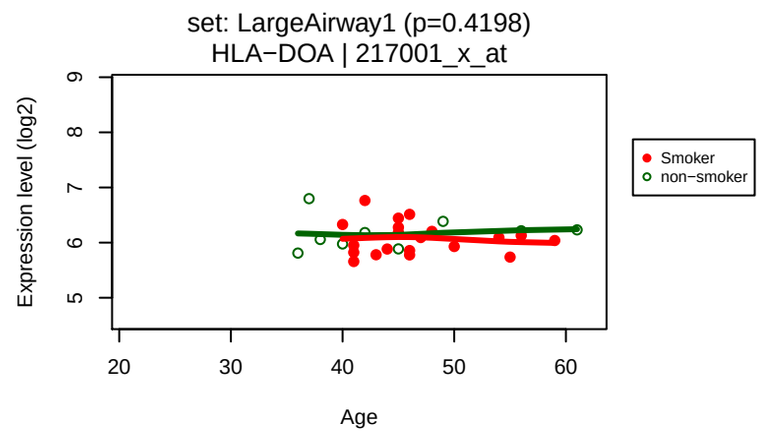

Gene expression of smokers and nonsmokers
gene: HLA−DOA | 217001_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0878
p=0.6928
p=0.6307
p=0.4198
p=0.8333
p=0.6171
p=0.5193
p=0.7287
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

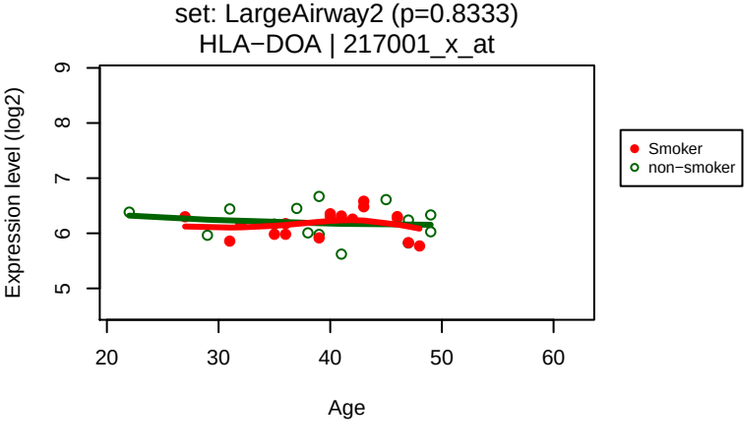

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DOA | 217001_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0878
p=0.6928
p=0.6307
p=0.4198
p=0.8333
p=0.6171
p=0.5193
p=0.7287
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

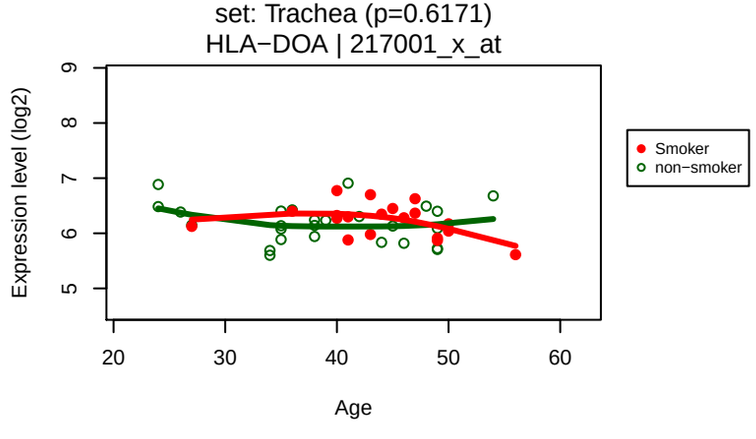

Gene expression ratio in Males
gene: HLA−DOA | 217001_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5281
p=0.1714
p=0.302
p=0.3239
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DOA | 217001_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4918
p=0.2325
p=0.9437
p=0.7243
SmallAirway2
LargeAirway2
Trachea
Macrophage2

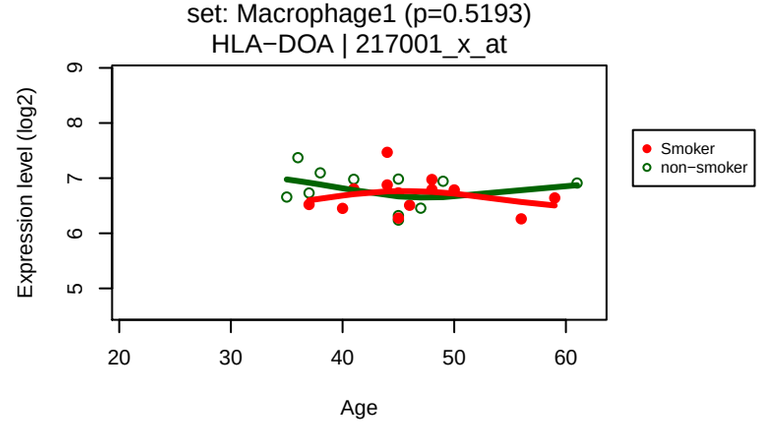

set: SmallAirway1 (p=0.0878)
HLA−DOA | 217001_x_at
Expression level (log2)
Age
Smoker
non−smoker

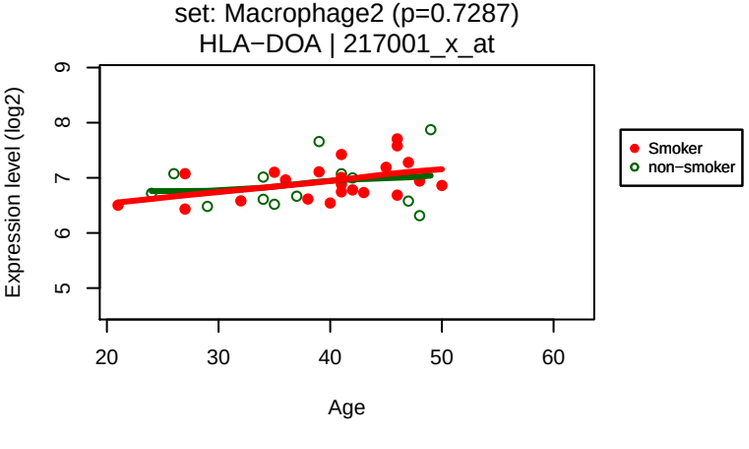

set: SmallAirway2 (p=0.6928)
HLA−DOA | 217001_x_at
Expression level (log2)
Age
Smoker
non−smoker

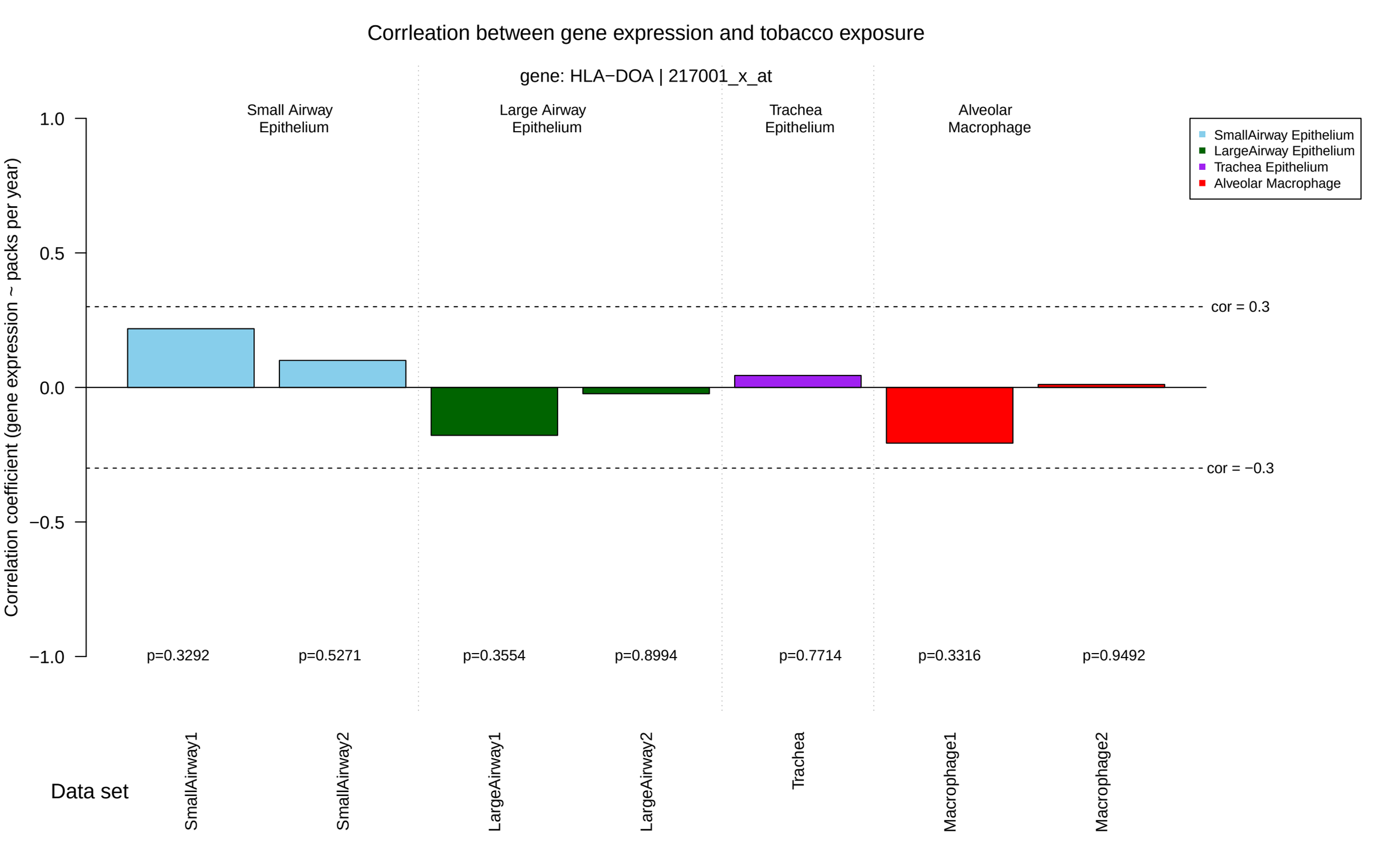

set: LargeAirway1 (p=0.4198)
HLA−DOA | 217001_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8333)
HLA−DOA | 217001_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6171)
HLA−DOA | 217001_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5193)
HLA−DOA | 217001_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7287)
HLA−DOA | 217001_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DOA | 217001_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3292
p=0.5271
p=0.3554
p=0.8994
p=0.7714
p=0.3316
p=0.9492
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2